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Price level determinacy without control of a monetary aggregate

A comment

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In his paper Woodford challenges a quantity theory of the price level, and in its place he argues for a fiscal theory of the price level. Woodford's principal argument for wanting us to think about the price level from a fiscal perspective is that quantity theorists have misled us along two important dimensions. First, quantity theorists have stressed the importance of a strict exogenous control over the stock of money, as an endogenous determination of the stock of money (such as under a monetary policy that holds constant the nominal interest rate) may lead to an indeterminate price level. As for the second issue, quantity theorists have stressed the importance of sharply distinguishing between monetary and nonmonetary assets, which has often led them to oppose technological advances that create close substitutes for money. Woodford challenges the view that it is important for a monetary authority, who is interested in a stable price level, to pursue these objectives. In my remarks I will first discuss the fiscal theory of the price level, and then I will address these two issues.

1 The fiscal theory of the price level

The fiscal theory of the price level can be summarized with reference to the following equation of Woodford's paper:

$$\frac{W_t}{p_t} = \sum_{s=t}^{\infty} \frac{(\tau_s - g_s) + \Delta_s m_s}{\Pi_{j=t}^{s-1}(1 + r_j^b)}. \tag{1}$$

In this equation, W is the nominal value of net government liabilities, p is the price level, τ and g are government real lump-sum taxes and spending,

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m is the real money base, $\Delta = (R^b - R^m)/R^m$ where R^b is the gross nominal return to bonds and R^m is the gross nominal return to fiat money, and r^b is the real interest rate. In equilibrium, this is the present-value government budget constraint. If fiscal policy chooses arbitrary sequences $\{\tau_t, g_t\}$, then this equation may only be satisfied by a unique price level. What is meant by this statement is that if the government arbitrarily chooses a new sequence of taxes and spending, say $\{\tau_t', g_t'\}$, then at an unchanged price level the government budget constraint may not hold. The new equilibrium price level is then uniquely determined such that this constraint holds. In this sense fiscal policy determines the price level. To make matters concrete, I will think of a fiscal theory of prices as one in which equation (1) holds, and where taxes and spending are exogenous. (Woodford refers to such a fiscal policy as non-Ricardian, as equation (1) does not hold for arbitrary values for the price level. A Ricardian fiscal policy is such that equation (1) holds for arbitrary values for the price level.) In this sense it seems more appropriate to refer to this theory as a fiscal regime for price-level determination, rather than a fiscal theory of the price level (which is the terminology used by Aiyagari and Gertler, 1985, in a closely related setup).

This discussion brings out quite sharply the possible interdependence of monetary and fiscal policy in determining the price level. As Woodford notes, however, the issue is not so much if a quantity equation relates the price level to the stock of money, but rather that the stock of money (and/or the interest rate) may be determined entirely by its fiscal implications. The quantity equation, or rather the demand for real money balances, is given by the following equation of Woodford's paper:

$$\frac{M_t}{p_t} = L(y_t - g_t, \Delta_t). \tag{2}$$

In this equation, M is the nominal supply of money balances that households find useful for facilitating transactions, and y is an endowment of income. Obtaining a price level that ensures that the government satisfies its budget constraint thus requires a stock of money and a nominal interest rate that are consistent with the demand for real money balances. In this sense, the fiscal theory of the price level requires that either money or the nominal interest rate, or both, respond to fiscal policy.

2 Uses of the fiscal theory of the price level

At this point I would like to draw a sharp distinction between two uses of these two equations. For the first use, you could think of asking if, for experiences of actual economies, it makes more sense to think of prices (and hence money and interest rates) as having been largely determined by their

fiscal implications, or if it makes more sense to think of money and interest rates as having been largely determined by some monetary policy that did not respond to fiscal policy. For the second use of these equations, you could think of asking, again for actual economies, if a policymaker interested in price stability would find any use of the ideas discussed by Woodford.

2.1 *Does the fiscal theory of the price level help explain the data?*

For interpreting the history of actual economies, I think these equations bring out that this paper is perhaps not so much about a fiscal theory of the price level, but rather about identifying monetary and fiscal policy. On this score, I do not think it is possible to identify monetary or fiscal policy by purely theoretical arguments. After all, this theory only says that money, prices, deficits, and real interest rates are related by the two above equations. For identifying monetary and fiscal policy, it tells you nothing else. Identifying policy strikes me as an empirical issue, and I no more believe that the money supply or interest rates have been purely fiscally determined than that the money supply or interest rates have been purely exogenously determined.

In fact, we have evidence supporting both arguments, although for different episodes. Sargent (1982), in his chapter "The Ends of Four Big Inflations," documents how the post-WWI episodes of hyperinflation in Austria, Hungary, Poland, and Germany were due to sudden large fiscal deficits and that these episodes were only resolved once the fiscal situation was resolved. I think of these episodes, which Woodford also mentions, as sharp illustrations of how fiscal policy has had first-order implications for the price level. On the other hand, Friedman and Schwartz (1963), in their book "A Monetary History of the U.S.," have documented many episodes in U.S. history where the Federal Reserve has exerted independent influence on the money supply (one would also like to show that during these episodes interest rates were not determined by fiscal policy).

Intermediate to these relatively clear-cut illustrations, many reduced-form statistical studies have attempted to document a relationship between deficits and money (such as Joines, 1985, who examined the period 1872–1983). It is not clear how these types of studies would have been able to correctly identify monetary policy, but at any rate many of these studies have found little statistical relation between peacetime deficits and money growth. I do not think of this issue as settled, though, and I think the theory developed by Woodford and others allows us to progress on this important matter.

2.2 *Does the fiscal theory of the price level tell us how to stabilize prices?*

Would a policymaker interested in price stability find Woodford's ideas useful, or should the policymaker follow the quantity-theory recommendation of

strict control of a monetary aggregate (along with recommending a Ricardian fiscal policy)? Let me consider, first, the quantity-theory recommendation. I think it is entirely legitimate to suppose there exists unobserved and inevitable technology shocks to the transaction services industry, shocks that lead to variations in the real demand for any particular monetary aggregate. How else do you explain postwar velocity movements in many of these aggregates? In this sense, attempting to exert control such that some monetary aggregate grows at a constant rate would not lead to stable prices, as it is difficult to find a monetary aggregate that, for both short and long periods of time, has been related to prices in a stable way. Moreover, since the technology shocks just mentioned are perhaps often unobserved, it is not even clear what behavior of some chosen aggregate would lead to stable prices. On this dimension, I think Woodford has a legitimate criticism of this quantity-theory recommendation.

How could these technology shocks be accommodated in a fiscal theory of the price level? At this point I think it is important to draw a distinction between real money balances that enter equation (1) and real money balances that enter equation (2). Equation (1) pertains to revenue accruing to the government, so the real money supply that enters this equation is the monetary base. Equation (2) pertains to money that facilitates transactions by the private sector, so the real money supply that enters this equation could be anything (such as the base, M1, M2, or some other nonreservable aggregate). Why is this important? Presumably the monetary authority can control the monetary base (perhaps even the real monetary base if there is enough substitution between the base and other aggregates), but as just argued it may have substantial difficulty in controlling the aggregate that enters equation (2). But it is the base that enters the fiscal theory of prices, so these technology shocks need not have any effect on prices.

Conceptually, then, the fiscal theory of the price level seems to offer some promise for designing a policy for price stability: if real interest rates are fairly stable, then exogenous and stable deficits would lead to stable prices. In particular, the stability of the price level would not depend on the stability of any money demand function.

There are two caveats to the above observation. First, as can be seen from equation (1), variations in the real interest rate would lead to variations in the price level. To see that this might be an important source of price-level instability, recall that in the U.S. ex-post real interest rates were negative (-1 percent) in the mid-70s, and quite high (8 percent) in the early to mid-80s. It may be possible, however, to have deficits respond to real interest rates so that the present value of deficits is fairly stable.

Second, the stability of prices under the fiscal theory, which Woodford alludes to, depends largely on the stability of budget deficits. Here I have

little confidence, especially without imposing the discipline that debt must be paid off by lower future spending or higher taxes, that deficits would behave in a stable way. In fact, if anything, I would expect that pursuit of a fiscal theory of prices (exogenous deficits) would lead to less stable prices than what we observe now, rather than more stable prices. This claim simply stems from observations on recent behavior of the deficit, and the substantial uncertainty as to how this debt might be paid off. Giving control of the price level to the same set of people who control the deficit does not strike me as a good idea. Clearly, though, fleshing out this notion would require much more modeling effort, as one would have to seriously consider how the behavior of deficits would change in response to this change in regime.

3 The pursuit of price stability

It thus seems arguable that neither the quantity-theory prescription nor the fiscal-theory prescription would lead to a stable price level. So what would? If one takes Woodford's model seriously, this question has an obvious answer. A Ricardian fiscal policy and a monetary policy that chooses a constant inflation rate will lead to an equilibrium with a constant inflation rate. The supply of money and the nominal interest rate are then both endogenous, but precisely so that the inflation rate is not. I think the question of price stability becomes interesting, and realistic, only if one supposes that the monetary authority does not contemporaneously observe the price level, and that it does not observe some shocks to the money demand function. How does the monetary authority accommodate shocks that it cannot observe?

To pursue price stability, two important features of monetary policy are that it does not require information which the monetary authority may not have, and it accommodates various shocks that would otherwise affect the price level. A constant money-growth rule surely satisfies the informational requirement, but unfortunately such a rule does not accommodate any variation in the demand for money. A constant interest-rate rule, however, is also simple to implement and would accommodate a variety of fluctuations in the demand for money. Given the real interest rate, the nominal interest rate determines expected inflation and thereby accommodates any expected variation in the demand for money. Despite this appeal, many have criticized such a monetary policy on the grounds that, under such a monetary policy, the price level at any point in time is indeterminate (see the references in the section of Woodford's paper that addresses this issue). Indeed, Woodford argues that the fiscal theory of the price level is more useful than the quantity theory as prices are always determined under the fiscal theory, even with a pegged nominal interest rate. This criticism of an interest-rate rule seems unwarranted: there is no reason why a monetary authority cannot maintain

a constant interest rate through credible announcements of expected growth rates in future supplies of money, and independently choose the current innovations in the supply of money to minimize any (expected) variation in the price level. Coleman (1995) shows how the price level can be determined in a Fisherian model in which the monetary authority pegs the nominal interest rate. Gilles (1994) and Coleman, Gilles, and Labadie (1995) provide a discussion of some of these issues in the context of recent liquidity-effect models.

This analysis as it relates to price-level stability, though, needs two obvious qualifications. First, variations in the real interest rate would lead to variations in inflation under a nominal interest-rate peg (recall that the fiscal theory suffers from the same criticism). As for the second qualification, it is not clear what the inflation process would look like in the face of unobserved and unanticipated shocks to the demand for money. Since these shocks are unanticipated, they are not accommodated by the constant interest-rate policy (although they would be accommodated by the fiscal theory of prices). Also, since these shocks are unobserved, they could not be accommodated by adjusting the supply of money. The success of this policy for stabilizing prices thus depends on the exact nature of these kinds of shocks. In terms of stabilizing prices, though, I think this problem is less severe than the corresponding one under a policy of maintaining strict exogenous control over some monetary aggregate. In this case prices would vary in response to any variation in the demand for money.

4 Summary

To summarize, I need to see more empirical work before I believe that the fiscal theory of the price level is useful for explaining how prices behaved in actual economies. In asking the right questions and providing a coherent framework to carry out such tests, Woodford's paper makes an important contribution. Quite apart from this issue, before I believe that we should behave in such a way as to determine prices from a fiscal-theory perspective, I need to see some simulations of the price level when realistic processes for taxes and spending are input into this kind of system. As I have argued here, such an experiment should be compared to one in which the nominal interest rate is pegged and the monetary authority influences the price level by attempting to choose a stable inflation process, where what is input here are shocks to the real interest rate and the technology in the transaction-services industry.

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